

Boundary Pressure as a Universal Driver Across Domains

A Cross-Domain Structural Principle within the TNA Framework

Abstract

We propose that boundary-induced indeterminacy generates a universal form of structural pressure that drives system evolution across domains. Within the TNA framework, boundary pressure F_B emerges whenever a system encounters limits of validity. We show that this mechanism appears in physics, machine learning, and cognition under structurally equivalent forms. This suggests that expansion under constraint is a domain-independent principle.

1. Introduction

Across disciplines, systems exhibit a common pattern:

- stable operation within a domain
- failure or divergence at limits
- subsequent structural adaptation

Traditionally, these are treated as domain-specific phenomena:

- singularities in physics
- generalization failure in machine learning
- paradoxes in cognition

We argue that these are instances of a single structural mechanism:

■ *boundary pressure*

2. TNA Formal Setting

A system is defined as:

$$\Sigma = (\Omega, \partial\Omega, R, H_s)$$

where:

- $\Omega \rightarrow$ domain of valid operation
- $\partial\Omega \rightarrow$ boundary of validity
- $R(x) \rightarrow$ reliability
- $H_s \rightarrow$ structural history

Definition (Boundary Indeterminacy)

$$I_B = \text{uncertainty or inconsistency localized at } \partial\Omega$$

Definition (Boundary Pressure)

$$F_B = f(I_B)$$

with:

$$F_B > 0 \quad \text{iff} \quad I_B > 0$$

3. Structural Law

$$\partial\Omega \Rightarrow I_B \Rightarrow F_B \Rightarrow \Omega^+$$

4. Domain Instantiations

4.1 Physics

System:

- classical mechanics

Boundary:

- high-curvature regimes

Manifestation:

- divergence from observation

Interpretation:

- $\partial\Omega \rightarrow$ breakdown of model
- $I_B \rightarrow$ inconsistency with empirical data
- $F_B \rightarrow$ need for new theory

Result:

$$\Omega \rightarrow \Omega^+ \quad (\text{e.g., new geometry})$$

4.2 Machine Learning

System:

- model trained on distribution D_1

Boundary:

- out-of-distribution inputs

Manifestation:

- sharp performance degradation

Interpretation:

- $\partial\Omega \rightarrow$ non-generalizable region

- $I_B \rightarrow$ prediction uncertainty / error
 - $F_B \rightarrow$ retraining or architecture change
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4.3 Cognition

System:

- coherent reasoning process

Boundary:

- paradoxes / existential questions

Manifestation:

- non-resolvable states

Interpretation:

- $\partial\Omega \rightarrow$ limits of conceptual structure
 - $I_B \rightarrow$ ambiguity / contradiction
 - $F_B \rightarrow$ generation of new concepts
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5. Structural Equivalence

Across all domains:

Failure Mode	$\rightarrow$	$\partial\Omega$
Uncertainty	$\rightarrow$	I_B
Adaptation Need	$\rightarrow$	F_B
Evolution	$\rightarrow$	Ω^+

6. Minimal Dynamical Rule

Expansion Rule

$$\begin{array}{l} \text{if } F_B > \tau: \\ \Omega \rightarrow \Omega^+ \end{array}$$

Persistence Rule

$$R(\Omega^+) \geq R(\Omega) \text{ locally}$$

7. Non-Termination

Proposition

If new boundaries emerge after expansion:

$$\partial\Omega^+ \neq \emptyset$$

then:

$$\Sigma \text{ evolves indefinitely}$$

8. Interpretation

Boundary pressure is not noise or failure.

It is:

| *the generative force that drives structural evolution*

9. Implications

Physics

- singularities are not endpoints
 - they are expansion triggers
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Machine Learning

- OOD failure is not error
 - it is structural signal
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Cognition

- paradox is not breakdown
 - it is concept formation
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10. Unified Principle

Systems evolve by confronting their limits

11. Relation to Previous Results

- Formal paper → shows limits of closure
 - Interpretation paper → maps circularity
 - This paper → introduces dynamics
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12. Limitations

- no explicit metric for F_B
 - domain-specific instantiations require refinement
 - empirical validation remains open
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13. Conclusion

We have shown that:

- boundary-induced indeterminacy generates pressure
 - pressure drives expansion
 - this mechanism is domain-independent
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14. Final Statement

■ *Boundary pressure is a universal driver of structural evolution across domains.*